

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-30

Contents

Action Center	2
Full Sample PC1	3
Position Sheet	4
Cluster Overview	6
Cluster Details	7
Long ETF Positions (full)	59
ETF Net Signal Strength	68
Diversifier Candidates	70
Run Parameters	71

Action Center**Run date:** 2026-05-30 **Universe:** 660 ETFs / 26 clusters**Full PC1:** LONG since 2026-04-13 (D+33, 5.2%) **Clusters long:** 14/26 **ETFs long:** 373/660 (57%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

BUY (2)

cluster	theme	spread_buy	p_h5	nss
cluster_5	Aggregate Fixed Income	0.058	0.67	0.521
cluster_25	International Fixed Income	0.036	0.54	-0.306

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_18	Consumer Staples	-0.549	0.96	-0.576

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_24	Cash / Ultra-Short Fixed Income	0.343	1	1.00	1.000	1.00
cluster_22	Japan Equity	8.737	1	1.00	1.000	0.99
cluster_2	U.S. Mid Cap Equity	5.255	1	1.00	1.000	0.96
cluster_14	Financials	2.245	1	1.00	0.902	0.65
cluster_11	High Yield Bond	0.766	1	0.99	0.764	0.55

Full Sample PC1

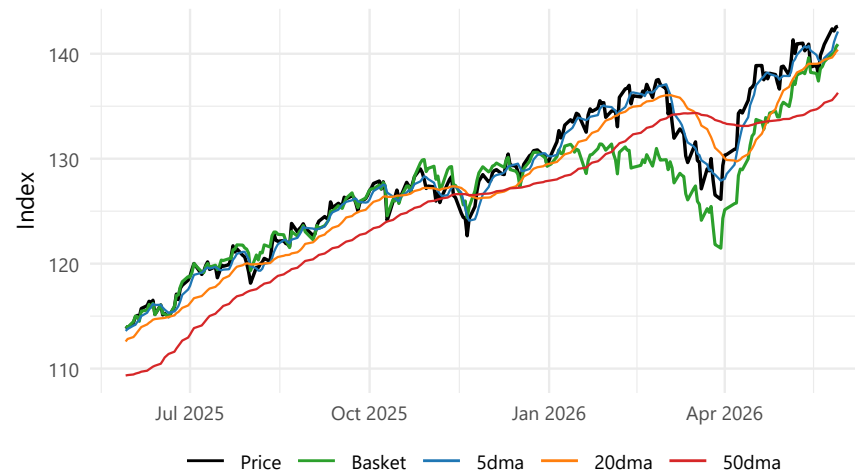


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-04-13 (33 days, 5.2%)

5dma: 142.121 20dma: 140.380 50dma: 136.282

Spread vs 50dma: +5.840 Spread vs 20dma: +1.742

P(buy crossover within 5 days): 100.0% P(sell): 1.2%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VTV	cluster_1	U.S. Large Cap Value	1.000	33	3.4%	7.90%	0.98
VOO	cluster_4	U.S. Large Cap Equity	1.000	33	7.9%	5.58%	0.99
VHT	cluster_12	Healthcare	1.000	16	1.6%	5.15%	0.93
VXF	cluster_7	U.S. Extended Market Equity	1.000	33	12.6%	3.88%	0.97
QQQ	cluster_6	U.S. Large Cap Growth	1.000	34	24.4%	3.68%	0.98
VEA	cluster_3	Developed ex-US Equity	0.995	34	3.2%	3.96%	0.99
SRLN	cluster_23	Bank Loan / Floating Rate	0.957	37	1.3%	6.22%	0.89
FLOT	cluster_23	Bank Loan / Floating Rate	0.957	37	1.3%	5.85%	0.89
BKLN	cluster_23	Bank Loan / Floating Rate	0.957	37	1.3%	4.18%	0.89
FLRN	cluster_23	Bank Loan / Floating Rate	0.957	37	1.3%	2.55%	0.89
FLTR	cluster_23	Bank Loan / Floating Rate	0.957	37	1.3%	0.69%	0.89
IEMG	cluster_8	Emerging Markets Equity	0.906	34	10.1%	3.12%	0.99
VNQ	cluster_15	Real Estate	0.835	34	3.4%	5.44%	0.99
MUB	cluster_16	Municipal Bond	0.653	1	0.1%	18.09%	0.95
BND	cluster_5	Aggregate Fixed Income	0.521	0	0.0%	17.93%	0.99
BBAX	cluster_9	Asia-Pacific Equity	0.266	4	0.7%	2.38%	0.97
IGF	cluster_9	Asia-Pacific Equity	0.266	4	0.7%	1.52%	0.97
EWZ	cluster_9	Asia-Pacific Equity	0.266	4	0.7%	0.49%	0.97
VTIP	cluster_21	Inflation-Protected Fixed I...	0.234	2	0.3%	34.46%	0.90
BWX	cluster_25	International Fixed Income	-0.306	0	0.0%	8.68%	0.93

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	0.648	1.097	0.96
IJH	cluster_2	U.S. Mid Cap Equity	1.00	0.955	1.028	0.99
EWJ	cluster_22	Japan Equity	1.00	0.994	0.908	0.96
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.419	0.84
SHV	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.342	0.84
MINT	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.177	0.84

GBIL	cluster_24	Cash / Ultra-Short Fixed In...	1.00	1.000	0.118	0.84
HYG	cluster_11	High Yield Bond	0.99	0.546	0.916	0.95
GUNR	cluster_10	Materials	0.95	0.442	0.773	0.96
AML	cluster_10	Materials	0.95	0.442	0.114	0.96
URA	cluster_10	Materials	0.95	0.442	0.081	0.96

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
1	cluster_1	U.S. Large Cap Value	U.S. Large Cap Va...	77	LONG		33	3.4%	1.000	+0.007
4	cluster_4	U.S. Large Cap Equity	U.S. Large Cap Eq...	58	LONG	shifted	33	7.9%	1.000	+0.005
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	45	LONG	shifted	34	24.4%	1.000	+1.128
7	cluster_7	U.S. Extended Market Equity	U.S. Extended Mar...	39	LONG	shifted	33	12.6%	1.000	+1.344
12	cluster_12	Healthcare	Healthcare	17	LONG	shifted	16	1.6%	1.000	+1.727
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT	shifted			1.000	+1.482
3	cluster_3	Developed ex-US Equity	Developed ex-US E...	61	LONG	shifted	34	3.2%	0.995	-0.005
22	cluster_22	Japan Equity	Japan Equity	9	FLAT	shifted			0.994	-0.005
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	LONG	shifted	37	1.3%	0.957	+0.882
2	cluster_2	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	70	FLAT	shifted			0.955	+0.037
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	LONG		34	10.1%	0.906	+0.088
15	cluster_15	Real Estate	Real Estate	14	LONG		34	3.4%	0.835	-0.068
16	cluster_16	Municipal Bond	Municipal Bond	13	LONG		1	0.1%	0.653	+0.675
14	cluster_14	Financials	Financials	14	FLAT				0.648	+0.052
11	cluster_11	High Yield Bond	High Yield Bond	21	FLAT	shifted			0.546	-0.454
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	58	LONG	shifted	0	0.0%	0.521	-0.477
10	cluster_10	Materials	Materials	21	FLAT	shifted			0.442	+0.255
9	cluster_9	Asia-Pacific Equity	Asia-Pacific Equi...	23	LONG	shifted	4	0.7%	0.266	-0.075
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	LONG	shifted	2	0.3%	0.234	-0.721
25	cluster_25	International Fixed Income	International Fix...	6	LONG	shifted	0	0.0%	-0.306	-1.305
17	cluster_17	Gold	Gold	12	FLAT				-0.574	+0.096
18	cluster_18	Consumer Staples	Consumer Staples	11	FLAT	shifted			-0.576	+0.162
13	cluster_13	China Equity	China Equity	15	FLAT	shifted			-0.795	-0.886
20	cluster_20	Broad Commodities	Broad Commodities	10	FLAT	shifted			-0.933	-1.926
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT	shifted			-0.993	-0.417
19	cluster_19	India Equity	India Equity	11	FLAT	shifted			-1.000	-0.730

Cluster Details

Cluster 1 – U.S. Large Cap Value

Members: 77 **Var explained:** 0.878 **Signal:** LONG since 2026-04-13 (33 days, 3.4%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +22.4%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +20.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	22	0.2846
U.S. Large Cap Value	10	0.1364
Value Equity	4	0.0527
U.S. Low Volatility Equity	4	0.0490
U.S. Multifactor Equity	3	0.0400
U.S. Value Factor Equity	3	0.0394
U.S. Mid Cap Equity	3	0.0394
Dividend Equity	3	0.0380

Utilities	3	0.0364
U.S. Fundamental Equity	2	0.0272
U.S. Mid Cap Value	2	0.0271
U.S. Equity	2	0.0270
U.S. Equal Weight Equity	2	0.0270
U.S. Small Cap Low Volatility	2	0.0247
U.S. Mega Cap Value	1	0.0135
U.S. Revenue Weighted Equity	1	0.0134
Large Cap Blend Equities	1	0.0133
U.S. Size Factor Equity	1	0.0132
U.S. Buyback Equity	1	0.0131
U.S. Large Cap Equity	1	0.0130
U.S. Quality / Factor Equity	1	0.0128
U.S. Mid Cap Low Volatility	1	0.0128
U.S. Defensive Equity	1	0.0125
Global Equity	1	0.0120
Communication Services	1	0.0110
Real Estate	1	0.0105

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
VTV	1.008	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
29	IWD	0.0137	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
32	JPUS	0.0137	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
37	VONV	0.0137	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
44	SCHV	0.0137	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
3	RSP	0.0136	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
14	PRF	0.0136	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
19	CFA	0.0136	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
21	NULV	0.0136	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
25	FNDX	0.0136	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
31	IVE	0.0136	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
35	VOOV	0.0136	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
41	ILCV	0.0136	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
43	IUSV	0.0136	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
45	SPYV	0.0136	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG

47	VTV	0.0136	Vanguard Value ETF	U.S. Large Cap Value	LONG
63	VOE	0.0136	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
68	DTD	0.0136	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
73	DGRO	0.0136	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
17	AIVL	0.0135	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
27	IWX	0.0135	iShares Russell Top 200 Value ETF	Value Equity	LONG
28	ONEV	0.0135	State Street SPDR Russell 1000 Low Volatility	U.S. Equity	FLAT
38	MGV	0.0135	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
40	VYM	0.0135	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
59	IWS	0.0135	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
67	DLN	0.0135	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
76	JPME	0.0135	JPMorgan Diversified Return U.S. Mid Cap	U.S. Equity	FLAT
15	RWL	0.0134	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Equity	LONG
48	FTA	0.0134	First Trust Large Cap Value AlphaDEX Fund	Value Equity	LONG
60	ONEY	0.0134	State Street SPDR Russell 1000 Yield Focus	U.S. Dividend Equity	LONG
65	PFM	0.0134	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
6	FEX	0.0133	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
26	EQAL	0.0133	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
4	FVD	0.0132	First Trust VL Dividend	U.S. Dividend Equity	LONG
22	SDY	0.0132	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
23	FDVV	0.0132	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
33	SIZE	0.0132	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
39	SCHD	0.0132	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
42	IMCB	0.0132	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
53	DVY	0.0132	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
74	VIG	0.0132	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
77	JVAL	0.0132	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
5	PKW	0.0131	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
30	IWR	0.0131	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
46	VO	0.0131	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
51	SDOG	0.0131	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
10	PWV	0.0130	Invesco Large Cap Value ETF	Value Equity	LONG
70	DIA	0.0130	State Street SPDR Dow Jones Industrial Average	U.S. Large Cap Equity	LONG
11	RDVY	0.0129	First Trust Rising Dividend Achievers ETF	U.S. Dividend Equity	FLAT
62	SPYD	0.0129	State Street SPDR Portfolio S&P 500 High Dividend	U.S. Dividend Equity	LONG
72	LGLV	0.0129	State Street SPDR US Large Cap Low Volatility	U.S. Low Volatility Equity	LONG
13	REGL	0.0128	ProShares S&P MidCap 400 Dividend Aristocrats	Dividend Equity	LONG
34	VLUE	0.0128	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
57	XMLV	0.0128	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatility	FLAT
64	FTCS	0.0128	First Trust Capital Strength ETF	U.S. Quality / Factor Equity	LONG
66	OUSA	0.0128	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
16	DHS	0.0127	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
20	CFO	0.0127	VictoryShares US 500 Enhanced Volatility	U.S. Multifactor Equity	FLAT
55	RPV	0.0127	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
61	SMMV	0.0127	iShares MSCI USA Small-Cap Min Vol Factor	U.S. Small Cap Low Volatility	LONG

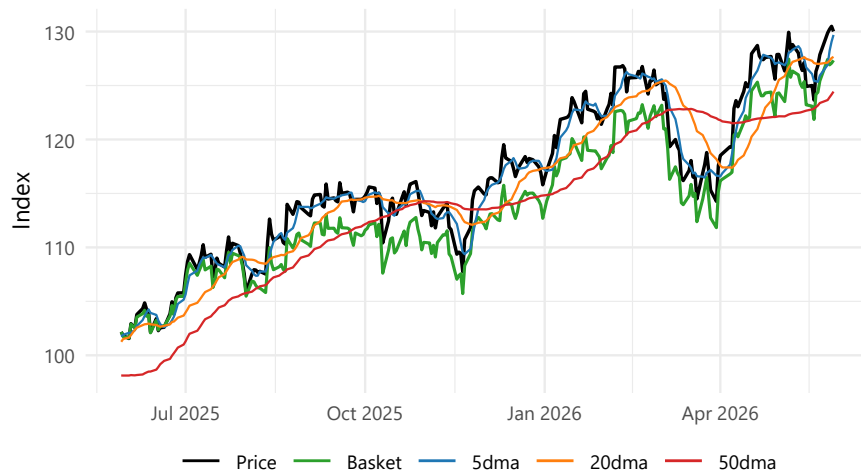
18	CDC	0.0125	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
71	USMV	0.0125	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
9	FIW	0.0124	First Trust Water ETF	Utilities	LONG
54	RDIV	0.0124	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
7	PHO	0.0123	Invesco Water Resources ETF	Utilities	FLAT
49	PEY	0.0123	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
56	SPHD	0.0123	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
12	FDL	0.0122	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
50	DIV	0.0122	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
2	ACWV	0.0120	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
24	LVHD	0.0120	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
58	XSLV	0.0120	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
36	HDV	0.0119	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
52	SMDV	0.0119	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
8	CGW	0.0117	Invesco S&P Global Water Index ETF	Utilities	FLAT
69	SPLV	0.0116	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
75	IYZ	0.0110	iShares U.S. Telecommunications ETF	Communication Services	FLAT
1	REM	0.0105	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 2 – U.S. Mid Cap Equity

Members: 70 **Var explained:** 0.903 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +41.3%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +27.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Industrials	11	0.1489
U.S. Small Cap Equity	10	0.1474
U.S. Mid Cap Equity	6	0.0888
U.S. Small Cap Value	6	0.0879
Consumer Discretionary	6	0.0780
U.S. Mid Cap Value	3	0.0443
U.S. Small Cap Growth	3	0.0441
U.S. Mid Cap Growth	3	0.0433
Leveraged / Inverse	2	0.0283
U.S. Small Cap Fundamental Equity	1	0.0149
Unknown	1	0.0149
U.S. Small Cap ESG Equity	1	0.0149
U.S. Mid/Small Cap Equity	1	0.0149
Mid Cap Value Equities	1	0.0148
U.S. Small/Mid Cap Fundamental Equity	1	0.0148
U.S. Small Cap Multifactor Equity	1	0.0148
Dividend Equity	1	0.0146
Small Cap Value Equities	1	0.0146
U.S. Small/Mid Cap Equity	1	0.0145
U.S. Mid Cap Dividend Equity	1	0.0145
U.S. ESG Equity	1	0.0144
U.S. Mid Cap Quality Equity	1	0.0143

Value Equity	1	0.0143
U.S. Small Cap Dividend Equity	1	0.0143
U.S. Equity	1	0.0142
U.S. Small Cap Industrials	1	0.0141
U.S. Small Cap Momentum Equity	1	0.0140
U.S. Micro Cap Equity	1	0.0137
U.S. Small Cap Consumer Discretionary	1	0.0135

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.008

ticker	rep_wt	name	category	etf_signal
IJH	1.028	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	FNX	0.0149	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
14	NUSC	0.0149	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
17	MDY	0.0149	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
24	IVOO	0.0149	Vanguard S&P Mid-Cap 400 ETF		FLAT
27	IJR	0.0149	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
28	IJH	0.0149	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
29	SPMD	0.0149	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
31	VB	0.0149	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
46	FNDA	0.0149	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
55	VBR	0.0149	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
15	PRFZ	0.0148	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
16	SMLF	0.0148	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
25	VI00	0.0148	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
30	SPSM	0.0148	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
32	SCHA	0.0148	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
42	EZM	0.0148	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
48	IJJ	0.0148	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
50	IVOV	0.0148	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
51	MDYV	0.0148	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
4	IWM	0.0147	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
7	JHMM	0.0147	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
10	SCHM	0.0147	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
12	FYX	0.0147	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
18	IJT	0.0147	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
20	VI0G	0.0147	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT

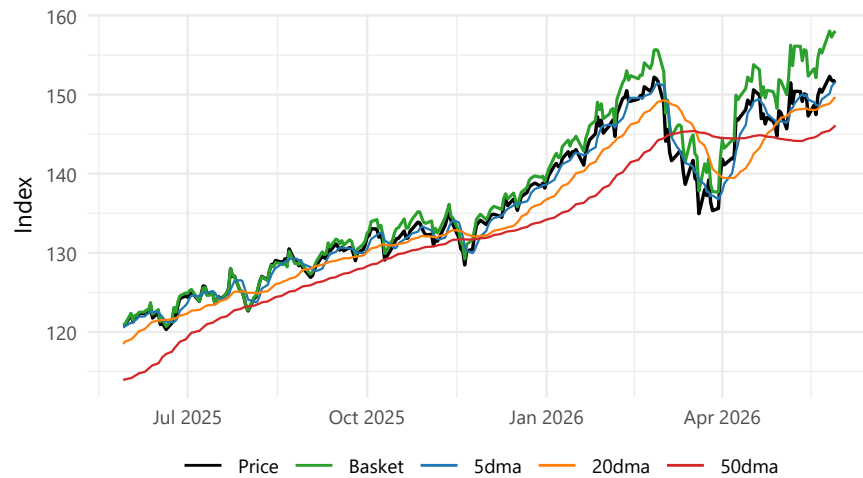
23	SLYG	0.0147	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
26	VTWO	0.0147	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
47	IWN	0.0147	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
63	RWK	0.0147	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
41	EES	0.0146	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
49	IJS	0.0146	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
52	SLYV	0.0146	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
54	VTWV	0.0146	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
69	SDVY	0.0146	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	FLAT
5	FXR	0.0145	First Trust Industrials AlphaDEX	Industrials	LONG
40	DON	0.0145	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
43	PSC	0.0145	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
53	VIOV	0.0145	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
57	RWJ	0.0145	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
70	ISMD	0.0145	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
19	IJK	0.0144	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
21	IVOG	0.0144	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
22	MDYG	0.0144	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
56	ETHO	0.0144	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
39	DES	0.0143	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
60	XMHQ	0.0143	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
62	RFV	0.0143	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
68	FYT	0.0143	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
64	OMFS	0.0142	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
58	PSCI	0.0141	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
6	IYJ	0.0140	iShares U.S. Industrials ETF	Industrials	FLAT
8	VIS	0.0140	Vanguard Industrials ETF	Industrials	FLAT
9	FIDU	0.0140	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
33	FXD	0.0140	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
59	PAVE	0.0140	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
65	RZV	0.0140	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
67	XSMO	0.0140	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
38	RSPD	0.0138	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
13	IWC	0.0137	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
1	XLI	0.0136	State Street Industrial Select Sector S...	Industrials	FLAT
35	FTXR	0.0136	First Trust Nasdaq Transportation ETF	Industrials	LONG
66	PSCD	0.0135	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
45	XTN	0.0133	State Street SPDR S&P Transportation ETF	Industrials	FLAT
61	PKB	0.0132	Invesco Building & Construction ETF	Industrials	LONG
2	XRT	0.0130	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
37	IYT	0.0130	iShares U.S. Transportation ETF	Industrials	FLAT
34	PEJ	0.0128	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
44	XHB	0.0127	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
3	JETS	0.0117	U.S. Global Jets ETF	Industrials	FLAT
36	ITB	0.0116	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 3 – Developed ex-US Equity

Members: 61 **Var explained:** 0.898 **Signal:** LONG since 2026-04-10 (34 days, 3.2%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +47.7%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +26.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)

**Category breakdown**

category	n_members	abs_weight
International Equity	19	0.3159
Developed ex-US Equity	9	0.1528
Global ex-US Equity	5	0.0849
Europe Equity	4	0.0674
International Dividend Equity	4	0.0660
Alternatives	4	0.0614
Eurozone Equity	3	0.0489
Commodity	1	0.0171
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162

Italy Equity	1	0.0159
United Kingdom Equity	1	0.0158
Europe Financials	1	0.0158
Netherlands Equity	1	0.0157
Sweden Equity	1	0.0156
Spain Equity	1	0.0150
Austria Equity	1	0.0147
Belgium Equity	1	0.0146
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
VEA	0.971	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0172	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
51	IEFA	0.0172	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	LONG
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	FLAT
40	GSIE	0.0171	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
6	IEV	0.0170	iShares Europe ETF	Europe Equity	FLAT
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	FLAT
19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
32	INTF	0.0170	iShares International Equity Factor ETF	International Equity	FLAT
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	FLAT
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
49	VGK	0.0170	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	LONG
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT

14	EZU	0.0167	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	LONG
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	LONG
26	JPIN	0.0167	JPMorgan Diversified Return Internation...	International Equity	FLAT
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	FLAT
38	VYMI	0.0167	Vanguard International High Dividend Yi...	International Dividend E...	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	LONG
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
34	IVLU	0.0166	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	LONG
31	IMTM	0.0164	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	LONG
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0162	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	LONG
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	LONG
15	IDV	0.0159	iShares International Select Dividend E...	International Dividend E...	LONG
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	LONG
13	EWU	0.0158	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
42	HDEF	0.0158	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
58	EWN	0.0157	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
27	DBEF	0.0156	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	LONG
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWP	0.0150	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0147	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWK	0.0146	iShares MSCI Belgium ETF	Belgium Equity	FLAT
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 4 – U.S. Large Cap Equity

Members: 58 **Var explained:** 0.904 **Signal:** LONG since 2026-04-13 (33 days, 7.9%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +25.1%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +24.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1626
U.S. Total Market Equity	7	0.1267
Global Equity	7	0.1243
Consumer Discretionary	6	0.0953
Asset Allocation	4	0.0659
U.S. Mega Cap Equity	3	0.0532
U.S. Quality / Factor Equity	3	0.0527
U.S. Multifactor Equity	3	0.0526
Communication Services	3	0.0475
U.S. ESG Equity	2	0.0360
U.S. Equity	2	0.0360
Commodity	1	0.0180
Large Cap Value Equities	1	0.0179
Large Cap Blend Equities	1	0.0179
Dividend Equity	1	0.0176

U.S. Dividend Equity	1	0.0172
U.S. Moat Equity	1	0.0166
Alternatives	1	0.0155
International IPO Equity	1	0.0139
Singapore Equity	1	0.0126

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
VOO	0.958	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	SPY	0.0181	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
31	IWV	0.0181	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
32	IYY	0.0181	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
37	ESGU	0.0181	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
38	IWB	0.0181	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
43	VTHR	0.0181	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
46	VONE	0.0181	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
48	VV	0.0181	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
49	ILCB	0.0181	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
50	ITOT	0.0181	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
51	IVV	0.0181	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
52	SCHB	0.0181	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
53	SCHX	0.0181	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
54	SPTM	0.0181	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
55	VOO	0.0181	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
56	VTI	0.0181	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
12	URTH	0.0180	iShares MSCI World ETF	Global Equity	LONG
27	JHML	0.0180	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
30	SUSA	0.0180	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
35	SPYX	0.0180	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
39	IWL	0.0180	iShares Russell Top 200 ETF	U.S. Equity	LONG
44	GSLC	0.0180	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
47	MGC	0.0180	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
7	ACWI	0.0179	iShares MSCI ACWI ETF	Global Equity	LONG
15	VT	0.0179	Vanguard Total World Stock Index ETF	Global Equity	LONG
23	TILT	0.0179	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
29	DSI	0.0179	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG

33	LRGF	0.0179	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
45	EPS	0.0179	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
13	CRBN	0.0178	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
14	SPGM	0.0178	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
34	OEF	0.0178	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
40	QUAL	0.0178	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
8	AOA	0.0177	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
25	QDF	0.0176	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
41	QUS	0.0176	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
24	I00	0.0175	iShares Global 100 ETF	Global Equity	LONG
26	CATH	0.0175	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
36	XLG	0.0175	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
42	SPHQ	0.0173	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0172	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
28	DGRW	0.0172	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
17	IYC	0.0169	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
58	OMFL	0.0168	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
22	MOAT	0.0166	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
18	VCR	0.0165	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
21	FDIS	0.0165	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
1	XLY	0.0163	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
10	AOM	0.0161	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
19	VOX	0.0160	Vanguard Communication Services ETF	Communication Services	LONG
20	FCOM	0.0159	Fidelity MSCI Communication Services In...	Communication Services	LONG
2	XLC	0.0156	State Street Communication Services Sel...	Communication Services	FLAT
5	QAI	0.0155	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
57	RTH	0.0153	VanEck Retail ETF	Consumer Discretionary	FLAT
9	AOK	0.0148	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
16	FPXI	0.0139	First Trust International Equity Opport...	International IPO Equity	LONG
3	ESPO	0.0137	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
6	EWS	0.0126	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 5 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.832 **Signal:** LONG since 2026-05-29 (0 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +5.5%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +5.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-13)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3283
Investment Grade Corporate Bond	13	0.2302
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0890
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0308
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.008

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
46	SCHR	0.0182	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
50	MBB	0.0180	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
52	SPMB	0.0179	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	LONG
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	LONG
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	LONG
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	LONG
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

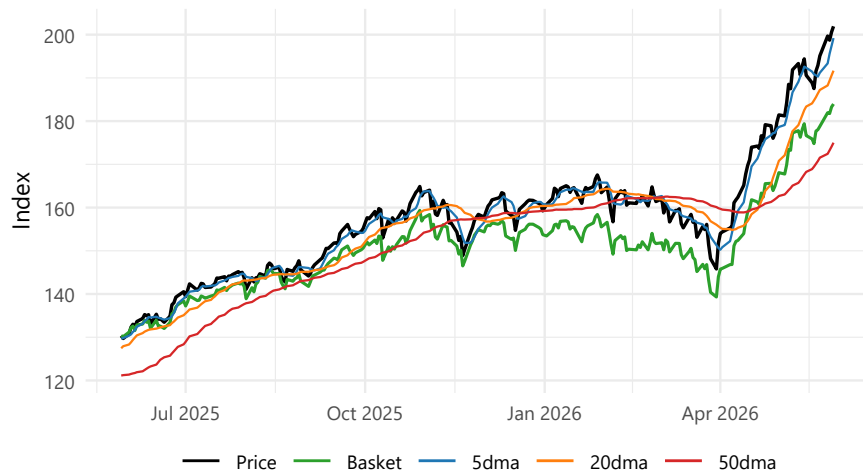
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	LONG
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	LONG
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	LONG
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0157	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	LONG
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	LONG
14	IBDR	0.0152	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0120	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 6 – U.S. Large Cap Growth

Members: 45 **Var explained:** 0.878 **Signal:** LONG since 2026-04-10 (34 days, 24.4%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +55.4%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +55.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Technology	23	0.5109
U.S. Large Cap Growth	11	0.2551
Growth Equity	2	0.0457
U.S. Equity	2	0.0455
U.S. Mega Cap Growth	1	0.0230
U.S. Equal Weight Equity	1	0.0226
Options Income	1	0.0209
U.S. Small Cap Technology	1	0.0209
Trend-Following Equity	1	0.0204
U.S. Trend-Following Equity	1	0.0178
Israel Equity	1	0.0172

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.020

ticker	rep_wt	name	category	etf_signal
QQQ	1.006	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	QQQ	0.0234	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
21	IGM	0.0234	iShares Expanded Tech Sector ETF	Technology	LONG
25	ONEQ	0.0234	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
29	VGT	0.0234	Vanguard Information Tech ETF	Technology	LONG
31	FTEC	0.0234	Fidelity MSCI Information Technology In...	Technology	LONG
23	IYW	0.0233	iShares U.S. Technology ETF	Technology	LONG
34	ILCG	0.0233	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
1	XLK	0.0232	State Street Technology Select Sector S...	Technology	LONG
22	IXN	0.0232	iShares Global Tech ETF	Technology	LONG
27	IWF	0.0232	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
32	VONG	0.0232	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
35	IUSG	0.0232	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
38	VUG	0.0232	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
17	XNTK	0.0231	State Street SPDR NYSE Technology ETF	Technology	LONG
24	NULG	0.0231	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
28	IVW	0.0231	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
30	VOOG	0.0231	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
36	SCHG	0.0231	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
37	SPYG	0.0231	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
11	QTEC	0.0230	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
26	IWY	0.0230	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
33	MGK	0.0230	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
6	AIQ	0.0228	Global X Artificial Intelligence & Tech...	Technology	LONG
16	FXL	0.0228	First Trust Technology AlphaDEX	Technology	LONG
20	PWB	0.0227	Invesco Large Cap Growth ETF	Growth Equity	LONG
10	QQEW	0.0226	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
14	XT	0.0225	iShares Future Exponential Technologies...	Technology	LONG
13	TDIV	0.0222	First Trust NASDAQ Technology Dividend ...	Technology	LONG
39	SPHB	0.0221	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
3	SMH	0.0220	VanEck Semiconductor ETF	Technology	LONG
2	SOXX	0.0219	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
45	QTUM	0.0218	Defiance Quantum ETF	Technology	LONG
5	ROBO	0.0217	ROBO Global Robotics and Automation Ind...	Technology	LONG
4	BOTZ	0.0216	Global X Robotics & Artificial Intellig...	Technology	LONG
12	PSI	0.0214	Invesco Semiconductors ETF	Technology	LONG
15	XSD	0.0214	State Street SPDR S&P Semiconductor ETF	Technology	LONG
43	FTXL	0.0214	First Trust Nasdaq Semiconductor ETF	Technology	LONG
8	NXTG	0.0213	First Trust Indxx NextG ETF	Technology	LONG
44	PTF	0.0211	Invesco Dorsey Wright Technology Moment...	Technology	LONG
19	QYLD	0.0209	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
42	PSCT	0.0209	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
18	PTNQ	0.0204	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
9	PRNT	0.0192	3D Printing (The) ETF	Technology	LONG

40	PTLC	0.0178	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
41	EIS	0.0172	iShares MSCI Israel ETF	Israel Equity	FLAT

Cluster 7 – U.S. Extended Market Equity

Members: 39 **Var explained:** 0.796 **Signal:** LONG since 2026-04-13 (33 days, 12.6%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.7%)
- Latest sell_cross: 2026-03-12
- Current 252d return: +26.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-12)



Category breakdown

category	n_members	abs_weight
Technology	12	0.2975
Industrials	4	0.0908
U.S. Mid Cap Growth	3	0.0841
U.S. Small Cap Growth	3	0.0833

U.S. Momentum Equity	3	0.0760
Growth Equity	2	0.0547
IPO Equity	2	0.0531
Convertible Bond	2	0.0530
Momentum Equity	2	0.0519
Financials	2	0.0513
U.S. Extended Market Equity	1	0.0281
U.S. Small Cap Momentum Equity	1	0.0256
U.S. Mid Cap Momentum Equity	1	0.0255
Consumer Discretionary	1	0.0253

Replication basket – VXF (cor=0.97) basket lead at lag 1: 0.019

ticker	rep_wt	name	category	etf_signal
VXF	0.998	Vanguard Extended Market ETF	U.S. Extended Market ...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
22	IWP	0.0281	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
30	VBK	0.0281	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
32	VXF	0.0281	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
25	VOT	0.0280	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
26	IMCG	0.0280	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
31	ISCG	0.0277	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
28	IWO	0.0276	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
29	VTWG	0.0275	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
18	FPX	0.0273	First Trust US Equity Opportunities ETF	IPO Equity	LONG
16	FTC	0.0272	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
20	RPG	0.0269	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
17	PDP	0.0268	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
19	CWB	0.0268	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
37	VFMO	0.0266	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
7	ARKQ	0.0262	ARK Autonomous Technology & Robotics ETF	Technology	LONG
23	ICVT	0.0262	iShares Convertible Bond ETF	Convertible Bond	LONG
15	FINX	0.0261	Global X FinTech ETF	Financials	FLAT
11	IPO	0.0258	Renaissance IPO ETF	IPO Equity	LONG
13	FDN	0.0257	First Trust DJ Internet Index Fund	Technology	LONG
2	SKYY	0.0256	First Trust Cloud Computing ETF	Technology	LONG
27	DWAS	0.0256	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
21	XMMO	0.0255	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG

5	ARKK	0.0254	ARK Innovation ETF	Technology	FLAT
6	ARKW	0.0254	ARK Next Generation Internet ETF	Technology	LONG
9	IBUY	0.0253	Amplify Online Retail ETF	Consumer Discretionary	LONG
12	PNQI	0.0253	Invesco Nasdaq Internet ETF	Technology	FLAT
8	FV	0.0251	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
14	IPAY	0.0251	Amplify Digital Payments ETF	Financials	FLAT
24	MTUM	0.0251	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
1	IGV	0.0247	iShares Expanded Tech-Software Sector E...	Technology	LONG
39	PRN	0.0245	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
10	CIBR	0.0242	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
36	QMOM	0.0242	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
4	HACK	0.0241	Amplify Cybersecurity ETF	Technology	LONG
3	BLOK	0.0234	Amplify Blockchain Technology ETF	Technology	LONG
35	XAR	0.0232	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
33	PPA	0.0222	Invesco Aerospace & Defense ETF	Industrials	LONG
34	ITA	0.0208	iShares U.S. Aerospace & Defense ETF	Industrials	LONG
38	BLCN	0.0205	Siren NexGen Economy ETF	Technology	LONG

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.874 **Signal:** LONG since 2026-04-10 (34 days, 10.1%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +44.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7725
Asia Equity	2	0.0813
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0358
Global Dividend Equity	1	0.0347
South Korea Equity	1	0.0340

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.026

ticker	rep_wt	name	category	etf_signal
IEMG	0.929	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG

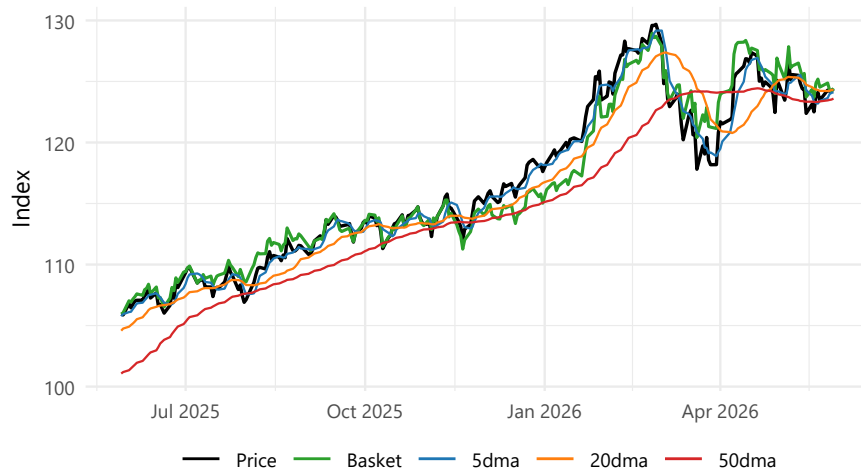
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	FLAT
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0397	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0391	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0386	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0358	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0347	Global X SuperDividend ETF	Global Dividend Equity	FLAT
25	FEMS	0.0346	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	LONG
1	EWY	0.0340	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 9 – Asia-Pacific Equity

Members: 23 **Var explained:** 0.685 **Signal:** LONG since 2026-05-22 (4 days, 0.7%)

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +17.2%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +17.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-22 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0979
International Dividend Equity	2	0.0961
International Equity	2	0.0958
Global Equity	2	0.0921
Real Estate	2	0.0917
Emerging Markets Fixed Income	2	0.0836
Currency	2	0.0813
Australia Equity	1	0.0486
Canada Equity	1	0.0462
Switzerland Equity	1	0.0431
South Africa Equity	1	0.0410
Latin America Equity	1	0.0394
New Zealand Equity	1	0.0394
Mexico Equity	1	0.0373
Brazil Equity	1	0.0333
Chile Equity	1	0.0330

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.001

ticker	rep_wt	name	category	etf_signal
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EWZ	0.094	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.290	iShares Global Infrastructure ETF	Global Equity	LONG
BBAX	0.453	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	EPP	0.0489	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
11	IDLV	0.0489	Invesco S&P International Developed Low...	International Equity	FLAT
23	BBAX	0.0489	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
4	EWA	0.0486	iShares MSCI Australia Index Fund	Australia Equity	LONG
10	DWX	0.0483	State Street SPDR S&P International Div...	International Dividend E...	LONG
3	PID	0.0478	Invesco International Dividend Achiever...	International Dividend E...	LONG
12	EFAV	0.0470	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
8	NFRA	0.0467	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
13	VNQI	0.0464	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
6	EWC	0.0462	iShares MSCI Canada Index Fund	Canada Equity	LONG
9	IGF	0.0454	iShares Global Infrastructure ETF	Global Equity	LONG
2	RWX	0.0453	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
5	EWL	0.0431	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
19	FXA	0.0431	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
14	EBND	0.0419	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
15	EMLC	0.0417	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
21	EZA	0.0410	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
18	ILF	0.0394	iShares Latin America 40 ETF	Latin America Equity	FLAT
20	ENZL	0.0394	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
22	FXC	0.0382	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
16	EWV	0.0373	iShares MSCI Mexico ETF	Mexico Equity	LONG
1	EWZ	0.0333	iShares MSCI Brazil ETF	Brazil Equity	FLAT
17	ECH	0.0330	iShares MSCI Chile ETF	Chile Equity	LONG

Cluster 10 – Materials

Members: 21 **Var explained:** 0.705 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +41.1% – filter **ALLOWS** new entries today

- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6531
MLP / Energy Infrastructure	4	0.1702
Energy	1	0.0501
Consumer Staples	1	0.0493
Norway Equity	1	0.0439
Industrial Metals	1	0.0333

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.042

ticker	rep_wt	name	category	etf_signal
AMLP	0.114	Alerian MLP ETF	MLP / Energy Infrastr...	FLAT
URA	0.081	Global X Uranium ETF	Materials	FLAT
GUNR	0.773	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

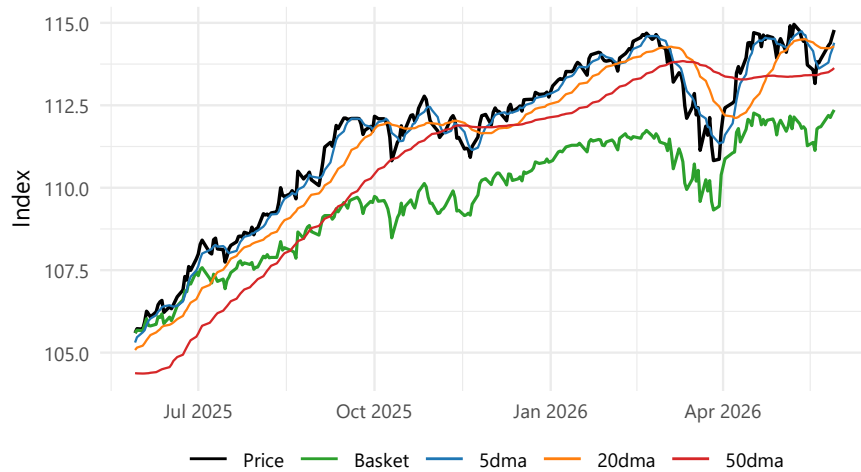
	ticker	PC_wt	name	category	etf_signal
15	GNR	0.0549	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0542	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0528	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0528	iShares U.S. Basic Materials ETF	Materials	LONG
21	NANR	0.0523	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0513	Vanguard Materials ETF	Materials	LONG
16	PICK	0.0508	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0506	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
1	XLB	0.0504	State Street Materials Select Sector SP...	Materials	LONG
20	IGE	0.0501	iShares North American Natural Resource...	Energy	FLAT
4	XME	0.0498	State Street SPDR S&P Metals & Mining E...	Materials	LONG
17	SLX	0.0496	VanEck Steel ETF	Materials	LONG
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0471	Global X Copper Miners ETF	Materials	LONG
5	EMLP	0.0441	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
9	MLPX	0.0439	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
18	NORW	0.0439	Global X MSCI Norway ETF	Norway Equity	LONG
7	AMLP	0.0417	Alerian MLP ETF	MLP / Energy Infrastruct...	FLAT
8	MLPA	0.0406	Global X MLP ETF	MLP / Energy Infrastruct...	FLAT
2	URA	0.0365	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0333	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 11 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +10.3%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +8.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5154
Preferred Stock	7	0.3063
Emerging Markets Fixed Income	4	0.1783

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.049

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0525	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0515	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0480	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWO	0.0466	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFFX	0.0463	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	LONG
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0434	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0369	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 12 – Healthcare

Members: 17 **Var explained:** 0.762 **Signal:** LONG since 2026-05-06 (16 days, 1.6%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +19.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
VHT	1.101	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0653	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0646	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0640	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	LONG

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0571	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0544	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	LONG
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0510	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.803 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +14.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.010

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

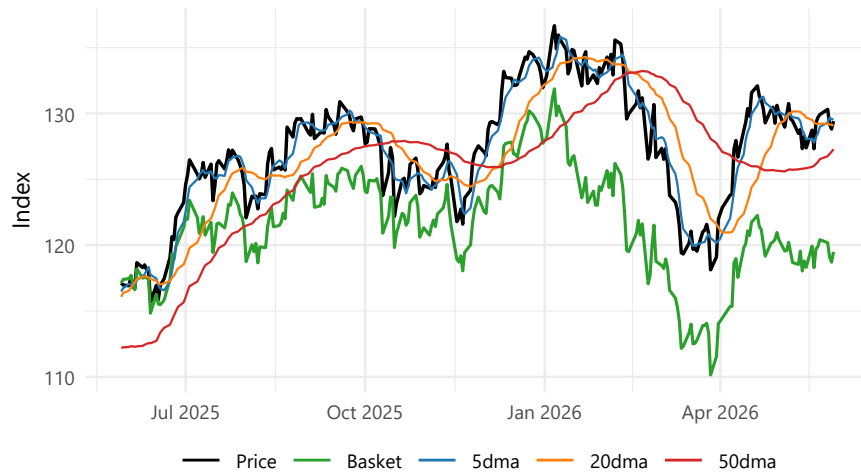
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0737	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	LONG
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0595	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
12	EWK	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0586	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – Financials

Members: 14 **Var explained:** 0.878 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +11.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.938
Global Equity	1	0.062

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.014

ticker	rep_wt	name	category	etf_signal
XLF	1.097	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0754	Vanguard Financials ETF	Financials	FLAT

11	FNCL	0.0754	Fidelity MSCI Financials Index ETF	Financials	LONG
5	FXO	0.0751	First Trust Financials AlphaDEX	Financials	FLAT
7	IYF	0.0751	iShares U.S. Financial ETF	Financials	FLAT
8	IYG	0.0747	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0740	State Street Financial Select Sector SP...	Financials	LONG
9	KBWB	0.0732	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0699	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0640	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0620	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (34 days, 3.4%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-03-10
- Current 252d return: +12.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG

1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	FLAT
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (1 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +6.2% – filter **ALLOWES** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.182

ticker	rep_wt	name	category	etf_signal
MUB	0.905	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	LONG
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0795	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	LONG
2	HYD	0.0747	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 17 – Gold

Members: 12 **Var explained:** 0.858 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +60.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1725
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.013

ticker	rep_wt	name	category	etf_signal
GLD	0.049	SPDR Gold Shares	Gold	FLAT
IAU	0.721	iShares Gold Trust Shares	Gold	FLAT
SLV	0.410	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT

1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0817	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0807	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.745 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +5.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-29 more recent than last effective buy 2026-05-22)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5731
Utilities	4	0.3523
U.S. Small Cap Consumer Staples	1	0.0747

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.587	State Street Consumer Staples Select ...	Consumer Staples	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	FLAT
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	FLAT
1	XLP	0.0974	State Street Consumer Staples Select Se...	Consumer Staples	FLAT
4	IYK	0.0948	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0945	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0898	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0891	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0879	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0879	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0874	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0747	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	FLAT

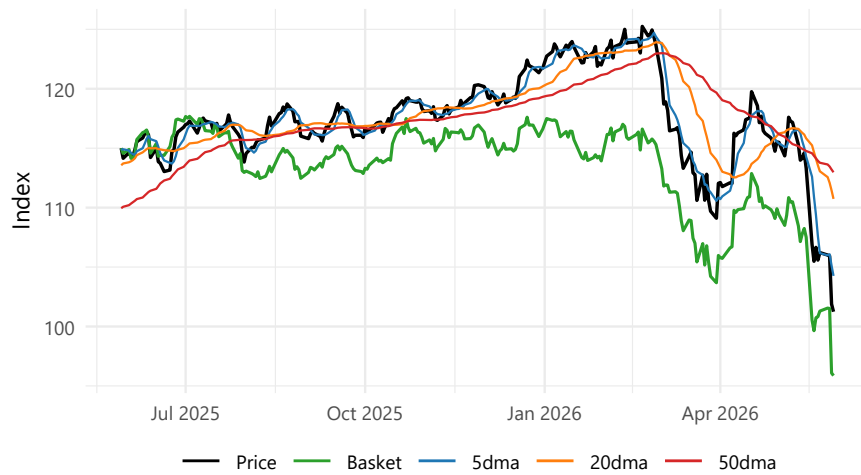
Cluster 19 – India Equity

Members: 11 **Var explained:** 0.370 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.3%)
- Latest sell_cross: 2026-05-13
- Current 252d return: -11.9% – filter **BLOCKS** new entries today

- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4642
Malaysia Equity	1	0.1127
Indonesia Equity	1	0.1061
Philippines Equity	1	0.1012
Vietnam Equity	1	0.0744
Turkey Equity	1	0.0598
U.S. Dividend Equity	1	0.0547
Platinum	1	0.0225
Natural Gas	1	0.0044

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.656	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.028	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1561	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1545	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1537	WisdomTree India Earnings Fund	India Equity	FLAT
9	EWM	0.1127	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
7	EIDO	0.1061	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
8	EPHE	0.1012	iShares MSCI Philippines ETF	Philippines Equity	FLAT
11	VNM	0.0744	VanEck Vietnam ETF	Vietnam Equity	FLAT
10	TUR	0.0598	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0547	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0225	abrdn Physical Platinum Shares ETF	Platinum	FLAT
6	UNG	0.0044	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – Broad Commodities

Members: 10 **Var explained:** 0.841 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-01-08 – filter **PASS** (252d at date = +2.8%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +48.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-27 more recent than last effective buy 2026-01-08)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6371
Oil	2	0.2056
Energy Commodities	1	0.1053
Agricultural Commodities	1	0.0520

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.075

ticker	rep_wt	name	category	etf_signal
PDBC	1.128	Invesco Optimum Yield Diversified Com...	Broad Commodities	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1091	Invesco DB Commodity Index Tracking Fund	Broad Commodities	FLAT
6	GSG	0.1091	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1090	Invesco Optimum Yield Diversified Commo...	Broad Commodities	FLAT
9	COMT	0.1089	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT

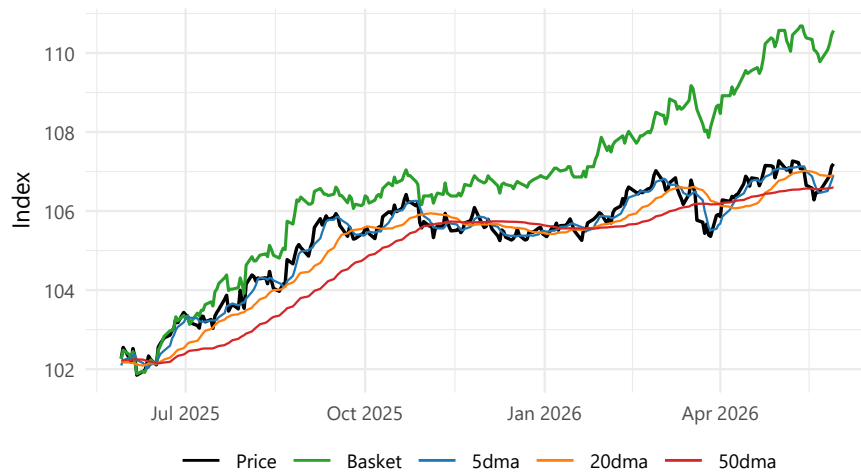
10	DBE	0.1053	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1033	United States Oil Fund	Oil	FLAT
5	DBO	0.1022	Invesco DB Oil Fund	Oil	FLAT
4	FTGC	0.1005	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
7	DJP	0.1005	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	DBA	0.0520	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.908 **Signal:** LONG since 2026-05-27 (2 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +5.1% – filter **ALLOWES** new entries today
- Decision: **LONG** (effective buy at 2026-05-27 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
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Inflation-Protected Fixed Income	10	1
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Replication basket – VTIP (cor=0.90) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
VTIP	1.723	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	LONG
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	LONG
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	LONG
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +40.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7799
Asia-Pacific Equity	2	0.2201

Replication basket – EWJ (cor=0.96) basket lead at lag 1: -0.037

ticker	rep_wt	name	category	etf_signal
EWJ	0.908	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
9	HEWJ	0.1116	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1115	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1107	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT

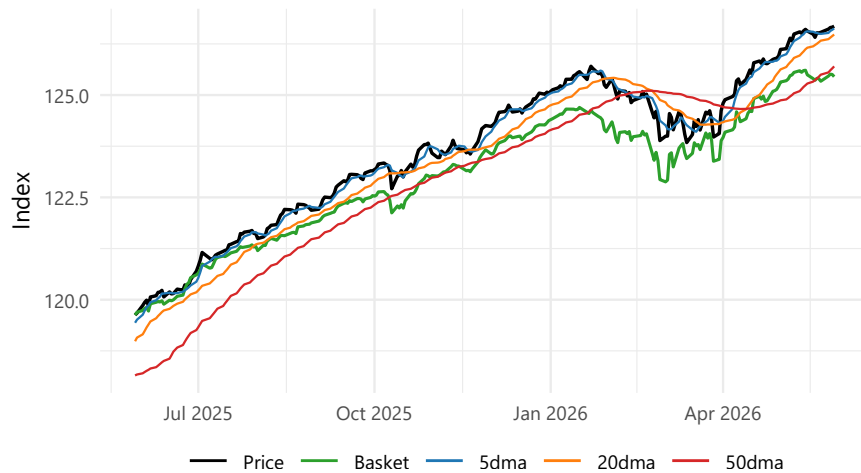
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1088	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.559 **Signal:** LONG since 2026-04-07 (37 days, 1.3%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-02-03
- Current 252d return: +6.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3630
Interest Rate Hedged Fixed Income	3	0.3245

Floating Rate Fixed Income	3	0.3125
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Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.209	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.311	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

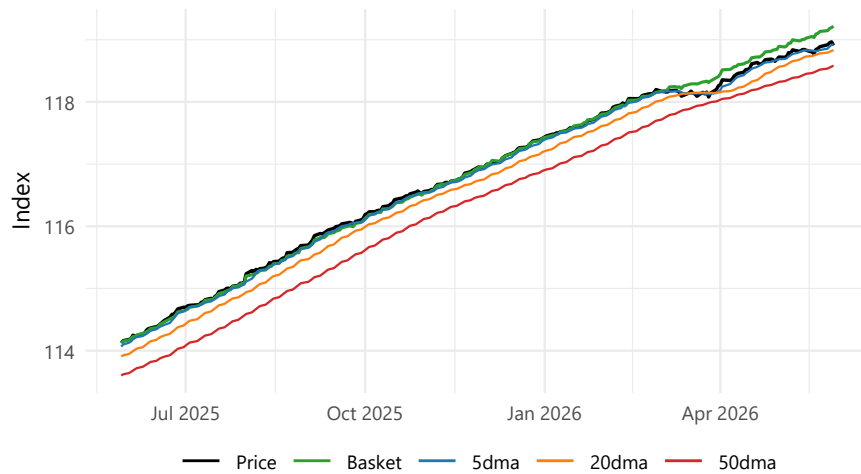
	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1213	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1168	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1082	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1058	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1042	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1036	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1007	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.448 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2022-10-31 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-03-24
- Current 252d return: +4.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-24 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.105

ticker	rep_wt	name	category	etf_signal
SHV	0.342	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.177	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.118	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.419	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	GSY	0.1339	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1268	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1263	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1250	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT

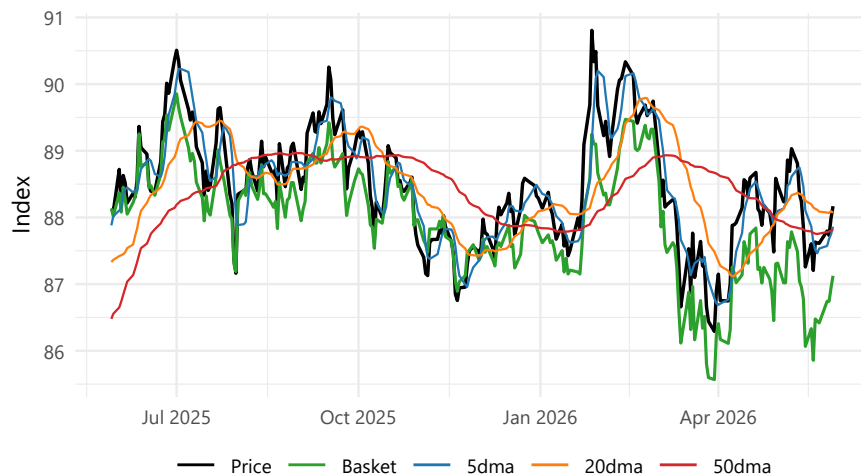
4	NEAR	0.1090	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1072	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.1040	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0884	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0793	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (0 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +0.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641

International Fixed Income	2	0.359
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Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.027

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

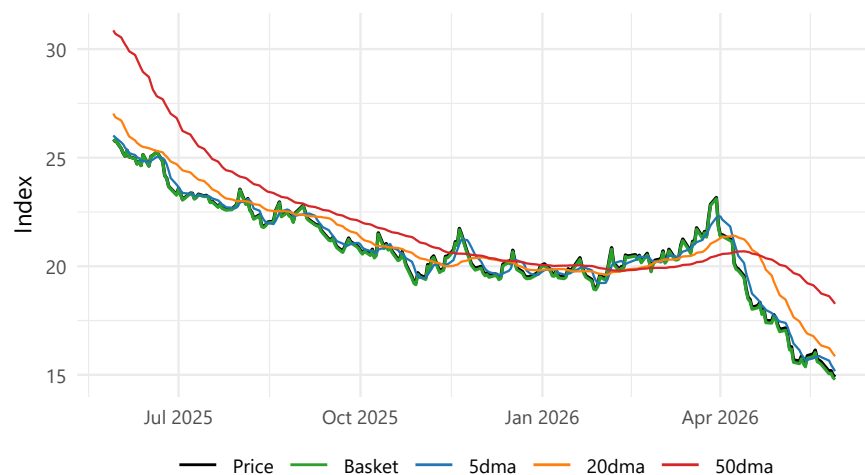
	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	LONG
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1648	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1451	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse

Members: 2 **Var explained:** 0.625 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-04-08
- Current 252d return: -42.4% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-08 more recent than last effective buy 2025-04-08)



Category breakdown

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.018

ticker	rep_wt	name	category	etf_signal
SQQQ	0.507	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
XLV	cluster_12	Healthcare	2026-05-21	5	0.9%	0.00	1.00
IYH	cluster_12	Healthcare	2026-05-20	6	1.8%	0.00	1.00
VHT	cluster_12	Healthcare	2026-05-19	7	2.0%	0.00	1.00
FHLC	cluster_12	Healthcare	2026-05-19	7	2.0%	0.00	1.00
IXJ	cluster_12	Healthcare	2026-05-22	4	-0.4%	0.00	1.00
PPH	cluster_12	Healthcare	2026-05-21	5	0.9%	0.00	1.00
RSPH	cluster_12	Healthcare	2026-05-21	5	1.3%	0.00	1.00
FXH	cluster_12	Healthcare	2026-04-10	34	2.9%	0.00	1.00
IBB	cluster_12	Healthcare	2026-05-28	1	-0.0%	0.09	0.77
XHE	cluster_12	Healthcare	2026-05-22	4	-2.0%	0.17	0.58
IHF	cluster_12	Healthcare	2026-05-08	14	0.6%	0.64	0.36
PSCH	cluster_12	Healthcare	2026-04-13	33	5.2%	0.95	0.05
BIL	cluster_24	Cash / Ultra-Short Fixe...	2022-05-19	1009	18.2%	0.00	1.00
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2022-08-18	947	17.7%	0.00	1.00
MINT	cluster_24	Cash / Ultra-Short Fixe...	2023-01-30	835	19.0%	0.00	1.00
VNLA	cluster_24	Cash / Ultra-Short Fixe...	2026-04-02	39	0.4%	0.00	1.00
QQEW	cluster_6	U.S. Large Cap Growth	2026-04-16	30	15.6%	0.00	1.00
PRNT	cluster_6	U.S. Large Cap Growth	2026-04-16	30	15.8%	0.00	1.00
NXTG	cluster_6	U.S. Large Cap Growth	2026-04-08	36	32.1%	0.00	1.00
XT	cluster_6	U.S. Large Cap Growth	2026-04-13	33	16.7%	0.00	1.00
QTUM	cluster_6	U.S. Large Cap Growth	2026-04-10	34	36.9%	0.00	1.00
XNTK	cluster_6	U.S. Large Cap Growth	2026-04-10	34	34.1%	0.00	1.00
QTEC	cluster_6	U.S. Large Cap Growth	2026-04-10	34	41.1%	0.00	1.00
TDIV	cluster_6	U.S. Large Cap Growth	2026-04-13	33	24.2%	0.00	1.00
FTEC	cluster_6	U.S. Large Cap Growth	2026-04-10	34	30.5%	0.00	1.00
PSCT	cluster_6	U.S. Large Cap Growth	2026-04-07	37	36.9%	0.00	1.00
VGT	cluster_6	U.S. Large Cap Growth	2026-04-10	34	30.3%	0.00	1.00
XLK	cluster_6	U.S. Large Cap Growth	2026-04-10	34	33.9%	0.00	1.00
IGM	cluster_6	U.S. Large Cap Growth	2026-04-09	35	30.4%	0.00	1.00
XSD	cluster_6	U.S. Large Cap Growth	2026-04-09	35	68.3%	0.00	1.00
AIQ	cluster_6	U.S. Large Cap Growth	2026-04-13	33	33.1%	0.00	1.00
IXN	cluster_6	U.S. Large Cap Growth	2026-04-10	34	33.3%	0.00	1.00
FXL	cluster_6	U.S. Large Cap Growth	2026-04-09	35	26.4%	0.00	1.00
PTLC	cluster_6	U.S. Large Cap Growth	2026-04-20	28	6.7%	0.00	1.00
PTNQ	cluster_6	U.S. Large Cap Growth	2026-04-17	29	13.9%	0.00	1.00
QQQ	cluster_6	U.S. Large Cap Growth	2026-04-10	34	20.8%	0.00	1.00
IYW	cluster_6	U.S. Large Cap Growth	2026-04-10	34	30.2%	0.00	1.00

SOXX	cluster_6	U.S. Large Cap Growth	2026-04-08	36	53.6%	0.00	1.00
VONG	cluster_6	U.S. Large Cap Growth	2026-04-14	32	11.1%	0.00	1.00
IWF	cluster_6	U.S. Large Cap Growth	2026-04-14	32	11.0%	0.00	1.00
PWB	cluster_6	U.S. Large Cap Growth	2026-04-10	34	18.2%	0.00	1.00
MGK	cluster_6	U.S. Large Cap Growth	2026-04-13	33	16.0%	0.00	1.00
IWY	cluster_6	U.S. Large Cap Growth	2026-04-14	32	11.5%	0.00	1.00
NULG	cluster_6	U.S. Large Cap Growth	2026-04-10	34	17.1%	0.00	1.00
SPHB	cluster_6	U.S. Large Cap Growth	2026-04-10	34	22.7%	0.00	1.00
FTXL	cluster_6	U.S. Large Cap Growth	2026-04-07	37	65.8%	0.00	1.00
VUG	cluster_6	U.S. Large Cap Growth	2026-04-13	33	15.0%	0.00	1.00
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	37	38.7%	0.00	1.00
SMH	cluster_6	U.S. Large Cap Growth	2026-04-08	36	41.6%	0.00	1.00
ONEQ	cluster_6	U.S. Large Cap Growth	2026-04-13	33	16.4%	0.01	0.99
SCHG	cluster_6	U.S. Large Cap Growth	2026-04-13	33	13.5%	0.01	0.99
ROBO	cluster_6	U.S. Large Cap Growth	2026-04-14	32	16.3%	0.01	0.99
ILCG	cluster_6	U.S. Large Cap Growth	2026-04-13	33	15.2%	0.01	0.99
IVW	cluster_6	U.S. Large Cap Growth	2026-04-13	33	14.9%	0.01	0.99
VOOG	cluster_6	U.S. Large Cap Growth	2026-04-13	33	14.9%	0.01	0.99
SPYG	cluster_6	U.S. Large Cap Growth	2026-04-13	33	14.9%	0.01	0.99
IUSG	cluster_6	U.S. Large Cap Growth	2026-04-13	33	14.5%	0.01	0.99
PSI	cluster_6	U.S. Large Cap Growth	2026-04-07	37	56.1%	0.02	0.98
BOTZ	cluster_6	U.S. Large Cap Growth	2026-04-16	30	9.4%	0.52	0.48
ILCV	cluster_1	U.S. Large Cap Value	2026-04-13	33	6.0%	0.00	1.00
MGV	cluster_1	U.S. Large Cap Value	2026-04-13	33	5.3%	0.00	1.00
FDVV	cluster_1	U.S. Large Cap Value	2026-04-14	32	6.6%	0.00	1.00
SDOG	cluster_1	U.S. Large Cap Value	2026-04-07	37	5.2%	0.00	1.00
USMV	cluster_1	U.S. Large Cap Value	2026-05-05	17	2.3%	0.00	1.00
VIG	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.8%	0.00	1.00
OUSA	cluster_1	U.S. Large Cap Value	2026-04-20	28	1.7%	0.00	1.00
DGRO	cluster_1	U.S. Large Cap Value	2026-04-13	33	3.6%	0.00	1.00
PFM	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.5%	0.00	1.00
SCHD	cluster_1	U.S. Large Cap Value	2026-04-21	27	4.9%	0.00	1.00
DLN	cluster_1	U.S. Large Cap Value	2026-04-13	33	4.7%	0.00	1.00
RWL	cluster_1	U.S. Large Cap Value	2026-04-10	34	7.5%	0.00	1.00
IVE	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.8%	0.00	1.00
VOOV	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.8%	0.00	1.00
SPYV	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.8%	0.00	1.00
VTV	cluster_1	U.S. Large Cap Value	2026-04-13	33	4.6%	0.00	1.00
DTD	cluster_1	U.S. Large Cap Value	2026-04-13	33	4.4%	0.00	1.00
RPV	cluster_1	U.S. Large Cap Value	2026-04-20	28	3.0%	0.00	1.00
SCHV	cluster_1	U.S. Large Cap Value	2026-04-10	34	6.6%	0.00	1.00
VLUE	cluster_1	U.S. Large Cap Value	2026-04-09	35	29.6%	0.00	1.00
RDIV	cluster_1	U.S. Large Cap Value	2026-04-21	27	4.7%	0.00	1.00
IUSV	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.6%	0.00	1.00
FNDX	cluster_1	U.S. Large Cap Value	2026-04-09	35	7.6%	0.00	1.00

SPYD	cluster_1	U.S. Large Cap Value	2026-04-21	27	2.8%	0.00	1.00
JVAL	cluster_1	U.S. Large Cap Value	2026-04-10	34	13.3%	0.00	1.00
DIA	cluster_1	U.S. Large Cap Value	2026-04-14	32	5.3%	0.00	1.00
RSP	cluster_1	U.S. Large Cap Value	2026-04-14	32	4.6%	0.00	1.00
PRF	cluster_1	U.S. Large Cap Value	2026-04-09	35	8.4%	0.00	1.00
AIVL	cluster_1	U.S. Large Cap Value	2026-04-10	34	5.0%	0.00	1.00
ACWV	cluster_1	U.S. Large Cap Value	2026-05-06	16	0.7%	0.00	1.00
IWX	cluster_1	U.S. Large Cap Value	2026-04-10	34	8.0%	0.00	1.00
SIZE	cluster_1	U.S. Large Cap Value	2026-04-13	33	5.9%	0.00	1.00
FEX	cluster_1	U.S. Large Cap Value	2026-04-09	35	6.5%	0.00	1.00
VYM	cluster_1	U.S. Large Cap Value	2026-04-10	34	4.7%	0.00	1.00
IMCB	cluster_1	U.S. Large Cap Value	2026-04-10	34	7.9%	0.00	1.00
IWD	cluster_1	U.S. Large Cap Value	2026-04-10	34	7.3%	0.00	1.00
VONV	cluster_1	U.S. Large Cap Value	2026-04-10	34	7.3%	0.00	1.00
PWV	cluster_1	U.S. Large Cap Value	2026-04-02	39	5.6%	0.00	1.00
ONEY	cluster_1	U.S. Large Cap Value	2026-04-10	34	4.5%	0.00	1.00
DHS	cluster_1	U.S. Large Cap Value	2026-04-20	28	1.5%	0.00	1.00
IWR	cluster_1	U.S. Large Cap Value	2026-04-10	34	6.7%	0.00	1.00
SDY	cluster_1	U.S. Large Cap Value	2026-05-20	6	0.8%	0.00	1.00
FTA	cluster_1	U.S. Large Cap Value	2026-05-21	5	1.3%	0.01	0.99
VO	cluster_1	U.S. Large Cap Value	2026-04-13	33	5.3%	0.01	0.99
NULV	cluster_1	U.S. Large Cap Value	2026-04-10	34	8.1%	0.01	0.99
FTCS	cluster_1	U.S. Large Cap Value	2026-05-20	6	-0.6%	0.01	0.99
HDV	cluster_1	U.S. Large Cap Value	2026-05-01	19	-0.5%	0.05	0.95
SPHD	cluster_1	U.S. Large Cap Value	2026-05-18	8	-0.0%	0.05	0.95
FVD	cluster_1	U.S. Large Cap Value	2026-05-22	4	-1.3%	0.02	0.93
FDL	cluster_1	U.S. Large Cap Value	2026-05-19	7	-1.0%	0.04	0.92
DIV	cluster_1	U.S. Large Cap Value	2026-04-07	37	0.8%	0.33	0.67
LVHD	cluster_1	U.S. Large Cap Value	2026-05-22	4	-1.9%	0.18	0.65
PKW	cluster_1	U.S. Large Cap Value	2026-05-20	6	0.5%	0.32	0.55
LGLV	cluster_1	U.S. Large Cap Value	2026-05-27	2	-1.0%	0.20	0.20
SMMV	cluster_1	U.S. Large Cap Value	2026-05-21	5	-0.5%	0.54	0.19
SPLV	cluster_1	U.S. Large Cap Value	2026-05-22	4	-2.5%	0.36	0.13
REGL	cluster_1	U.S. Large Cap Value	2026-05-26	3	-1.5%	0.56	-0.02
FIW	cluster_1	U.S. Large Cap Value	2026-05-08	14	-1.7%	0.59	-0.54
CIBR	cluster_7	U.S. Extended Market Eq...	2026-04-17	29	33.9%	0.00	1.00
HACK	cluster_7	U.S. Extended Market Eq...	2026-04-17	29	26.4%	0.00	1.00
IGV	cluster_7	U.S. Extended Market Eq...	2026-04-21	27	17.3%	0.00	1.00
FV	cluster_7	U.S. Extended Market Eq...	2026-04-14	32	12.7%	0.00	1.00
XAR	cluster_7	U.S. Extended Market Eq...	2026-05-22	4	6.2%	0.00	1.00
PPA	cluster_7	U.S. Extended Market Eq...	2026-05-27	2	1.8%	0.00	1.00
ITA	cluster_7	U.S. Extended Market Eq...	2026-05-26	3	2.6%	0.00	1.00
IPO	cluster_7	U.S. Extended Market Eq...	2026-04-13	33	22.5%	0.00	1.00
IWP	cluster_7	U.S. Extended Market Eq...	2026-04-15	31	5.6%	0.00	1.00
FDN	cluster_7	U.S. Extended Market Eq...	2026-04-10	34	16.7%	0.00	1.00

IMCG	cluster_7	U.S. Extended Market Eq...	2026-04-09	35	12.4%	0.00	1.00
ARKQ	cluster_7	U.S. Extended Market Eq...	2026-04-14	32	15.7%	0.00	1.00
SKYY	cluster_7	U.S. Extended Market Eq...	2026-04-17	29	23.4%	0.00	1.00
FTC	cluster_7	U.S. Extended Market Eq...	2026-04-10	34	12.2%	0.00	1.00
VBK	cluster_7	U.S. Extended Market Eq...	2026-04-10	34	12.6%	0.00	1.00
MTUM	cluster_7	U.S. Extended Market Eq...	2026-04-08	36	21.6%	0.00	1.00
RPG	cluster_7	U.S. Extended Market Eq...	2026-04-09	35	17.9%	0.01	0.99
FPX	cluster_7	U.S. Extended Market Eq...	2026-04-08	36	15.0%	0.01	0.99
ICVT	cluster_7	U.S. Extended Market Eq...	2026-04-08	36	12.5%	0.01	0.99
VOT	cluster_7	U.S. Extended Market Eq...	2026-04-14	32	7.9%	0.01	0.99
CWB	cluster_7	U.S. Extended Market Eq...	2026-04-08	36	12.3%	0.01	0.99
VTWG	cluster_7	U.S. Extended Market Eq...	2026-04-13	33	12.5%	0.02	0.98
IWO	cluster_7	U.S. Extended Market Eq...	2026-04-13	33	12.4%	0.02	0.98
BLOK	cluster_7	U.S. Extended Market Eq...	2026-04-13	33	22.0%	0.03	0.97
VFMO	cluster_7	U.S. Extended Market Eq...	2026-04-08	36	11.0%	0.06	0.94
ARKW	cluster_7	U.S. Extended Market Eq...	2026-04-14	32	16.8%	0.08	0.92
XMMO	cluster_7	U.S. Extended Market Eq...	2026-04-06	38	14.5%	0.08	0.92
PDP	cluster_7	U.S. Extended Market Eq...	2026-04-08	36	10.4%	0.10	0.90
BLCN	cluster_7	U.S. Extended Market Eq...	2026-04-15	31	14.2%	0.27	0.73
IBUY	cluster_7	U.S. Extended Market Eq...	2026-05-27	2	0.6%	0.16	0.62
QUS	cluster_4	U.S. Large Cap Equity	2026-04-14	32	4.8%	0.00	1.00
SPHQ	cluster_4	U.S. Large Cap Equity	2026-04-10	34	8.4%	0.00	1.00
LRGF	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.2%	0.00	1.00
QDF	cluster_4	U.S. Large Cap Equity	2026-04-13	33	7.9%	0.00	1.00
QUAL	cluster_4	U.S. Large Cap Equity	2026-04-14	32	6.0%	0.00	1.00
OMFL	cluster_4	U.S. Large Cap Equity	2026-04-10	34	9.2%	0.00	1.00
GSLC	cluster_4	U.S. Large Cap Equity	2026-04-13	33	8.9%	0.00	1.00
EPS	cluster_4	U.S. Large Cap Equity	2026-04-10	34	11.0%	0.00	1.00
ESGU	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.9%	0.00	1.00
SUSA	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.9%	0.00	1.00
JHML	cluster_4	U.S. Large Cap Equity	2026-04-13	33	8.1%	0.00	1.00
FPXI	cluster_4	U.S. Large Cap Equity	2026-04-10	34	14.2%	0.00	1.00
VONE	cluster_4	U.S. Large Cap Equity	2026-04-13	33	9.8%	0.00	1.00
VV	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.4%	0.00	1.00
ILCB	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.5%	0.00	1.00
IWB	cluster_4	U.S. Large Cap Equity	2026-04-13	33	9.8%	0.00	1.00
SCHX	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.1%	0.00	1.00
VTI	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.0%	0.00	1.00
ITOT	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.1%	0.00	1.00
SCHB	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.0%	0.00	1.00
IYY	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.0%	0.00	1.00
IWV	cluster_4	U.S. Large Cap Equity	2026-04-13	33	9.7%	0.00	1.00
VTHR	cluster_4	U.S. Large Cap Equity	2026-04-13	33	9.9%	0.00	1.00
VOO	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.3%	0.00	1.00
IVV	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.2%	0.00	1.00

SPYX	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.5%	0.00	1.00
IWL	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.9%	0.00	1.00
SPY	cluster_4	U.S. Large Cap Equity	2026-04-13	33	10.3%	0.00	1.00
MGC	cluster_4	U.S. Large Cap Equity	2026-04-13	33	11.5%	0.00	1.00
URTH	cluster_4	U.S. Large Cap Equity	2026-04-13	33	7.9%	0.00	1.00
SPTM	cluster_4	U.S. Large Cap Equity	2026-04-13	33	9.8%	0.00	1.00
TILT	cluster_4	U.S. Large Cap Equity	2026-04-13	33	8.4%	0.00	1.00
EWS	cluster_4	U.S. Large Cap Equity	2026-04-07	37	3.9%	0.00	1.00
DGRW	cluster_4	U.S. Large Cap Equity	2026-04-14	32	5.5%	0.00	1.00
OEF	cluster_4	U.S. Large Cap Equity	2026-04-13	33	12.3%	0.00	1.00
SPGM	cluster_4	U.S. Large Cap Equity	2026-04-10	34	9.1%	0.00	1.00
ACWI	cluster_4	U.S. Large Cap Equity	2026-04-13	33	8.3%	0.00	1.00
CRBN	cluster_4	U.S. Large Cap Equity	2026-04-13	33	8.4%	0.00	1.00
VT	cluster_4	U.S. Large Cap Equity	2026-04-13	33	8.0%	0.00	1.00
AOA	cluster_4	U.S. Large Cap Equity	2026-04-13	33	6.3%	0.01	0.99
IOO	cluster_4	U.S. Large Cap Equity	2026-04-13	33	11.4%	0.01	0.99
AOR	cluster_4	U.S. Large Cap Equity	2026-04-13	33	4.8%	0.01	0.99
CATH	cluster_4	U.S. Large Cap Equity	2026-04-13	33	9.3%	0.01	0.99
AOM	cluster_4	U.S. Large Cap Equity	2026-04-13	33	3.2%	0.02	0.98
XLG	cluster_4	U.S. Large Cap Equity	2026-04-13	33	12.0%	0.02	0.98
AOK	cluster_4	U.S. Large Cap Equity	2026-04-13	33	2.4%	0.03	0.97
QAI	cluster_4	U.S. Large Cap Equity	2026-04-08	36	4.5%	0.03	0.97
DSI	cluster_4	U.S. Large Cap Equity	2026-04-14	32	10.3%	0.04	0.96
IYC	cluster_4	U.S. Large Cap Equity	2026-04-14	32	0.3%	0.07	0.93
VOX	cluster_4	U.S. Large Cap Equity	2026-04-13	33	3.7%	0.29	0.71
FCOM	cluster_4	U.S. Large Cap Equity	2026-04-13	33	3.4%	0.30	0.70
VIDI	cluster_3	Developed ex-US Equity	2026-04-09	35	9.6%	0.00	1.00
IHDG	cluster_3	Developed ex-US Equity	2026-05-06	16	1.0%	0.00	1.00
HFXI	cluster_3	Developed ex-US Equity	2026-04-09	35	6.9%	0.00	1.00
EWN	cluster_3	Developed ex-US Equity	2026-04-10	34	7.4%	0.00	1.00
EWO	cluster_3	Developed ex-US Equity	2026-04-10	34	7.2%	0.00	1.00
PXF	cluster_3	Developed ex-US Equity	2026-04-08	36	6.0%	0.00	1.00
FNDF	cluster_3	Developed ex-US Equity	2026-04-08	36	5.9%	0.00	1.00
EWD	cluster_3	Developed ex-US Equity	2026-05-21	5	1.0%	0.00	1.00
VIGI	cluster_3	Developed ex-US Equity	2026-04-14	32	1.9%	0.00	1.00
EWG	cluster_3	Developed ex-US Equity	2026-04-14	32	2.9%	0.00	1.00
SCHF	cluster_3	Developed ex-US Equity	2026-04-10	34	6.2%	0.00	1.00
SPDW	cluster_3	Developed ex-US Equity	2026-04-10	34	6.1%	0.00	1.00
VEA	cluster_3	Developed ex-US Equity	2026-04-10	34	5.9%	0.00	1.00
CWI	cluster_3	Developed ex-US Equity	2026-04-10	34	5.8%	0.01	0.99
ACWX	cluster_3	Developed ex-US Equity	2026-04-10	34	5.8%	0.01	0.99
SCZ	cluster_3	Developed ex-US Equity	2026-04-10	34	4.4%	0.02	0.98
VYMI	cluster_3	Developed ex-US Equity	2026-04-09	35	1.9%	0.02	0.98
VEU	cluster_3	Developed ex-US Equity	2026-04-10	34	6.0%	0.02	0.98
IXUS	cluster_3	Developed ex-US Equity	2026-04-10	34	5.9%	0.02	0.98

FEP	cluster_3	Developed ex-US Equity	2026-04-09	35	3.1%	0.02	0.98
VXUS	cluster_3	Developed ex-US Equity	2026-04-10	34	5.9%	0.03	0.97
IQDF	cluster_3	Developed ex-US Equity	2026-04-10	34	5.0%	0.03	0.97
IMTM	cluster_3	Developed ex-US Equity	2026-04-10	34	2.4%	0.09	0.91
EWI	cluster_3	Developed ex-US Equity	2026-04-08	36	4.9%	0.10	0.90
FNDC	cluster_3	Developed ex-US Equity	2026-04-10	34	3.2%	0.21	0.79
DLS	cluster_3	Developed ex-US Equity	2026-04-13	33	1.9%	0.23	0.77
IDV	cluster_3	Developed ex-US Equity	2026-04-06	38	3.5%	0.30	0.70
HDEF	cluster_3	Developed ex-US Equity	2026-04-07	37	0.2%	0.36	0.58
IPKW	cluster_3	Developed ex-US Equity	2026-04-09	35	1.5%	0.43	0.57
HEWJ	cluster_22	Japan Equity	2026-04-08	36	4.7%	0.00	1.00
DBJP	cluster_22	Japan Equity	2026-04-08	36	4.7%	0.00	1.00
DXJ	cluster_22	Japan Equity	2026-05-01	19	6.3%	0.01	0.99
EWJV	cluster_22	Japan Equity	2026-05-06	16	2.2%	0.01	0.99
VPL	cluster_22	Japan Equity	2026-04-10	34	11.7%	0.02	0.98
EWJ	cluster_22	Japan Equity	2026-04-13	33	5.2%	0.04	0.96
BBJP	cluster_22	Japan Equity	2026-04-30	20	4.2%	0.05	0.95
LQDH	cluster_23	Bank Loan / Floating Ra...	2026-04-06	38	2.0%	0.00	1.00
IGHG	cluster_23	Bank Loan / Floating Ra...	2026-04-07	37	1.8%	0.00	1.00
IGBH	cluster_23	Bank Loan / Floating Ra...	2026-04-08	36	2.5%	0.00	1.00
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-04-07	37	1.8%	0.84	0.16
FTXR	cluster_2	U.S. Mid Cap Equity	2026-04-17	29	8.0%	0.00	1.00
ISMD	cluster_2	U.S. Mid Cap Equity	2026-04-09	35	11.4%	0.00	1.00
PEJ	cluster_2	U.S. Mid Cap Equity	2026-05-26	3	1.9%	0.00	1.00
SCHA	cluster_2	U.S. Mid Cap Equity	2026-04-09	35	10.5%	0.02	0.98
SCHM	cluster_2	U.S. Mid Cap Equity	2026-04-09	35	8.8%	0.02	0.98
FXD	cluster_2	U.S. Mid Cap Equity	2026-05-27	2	0.2%	0.00	0.96
IVOG	cluster_2	U.S. Mid Cap Equity	2026-04-09	35	7.9%	0.04	0.96
MDYG	cluster_2	U.S. Mid Cap Equity	2026-04-09	35	8.1%	0.04	0.96
IJK	cluster_2	U.S. Mid Cap Equity	2026-04-09	35	8.0%	0.05	0.95
IJS	cluster_2	U.S. Mid Cap Equity	2026-04-10	34	6.4%	0.06	0.94
PSCD	cluster_2	U.S. Mid Cap Equity	2026-05-27	2	-0.5%	0.06	0.80
FXR	cluster_2	U.S. Mid Cap Equity	2026-05-26	3	0.5%	0.18	0.79
RSPD	cluster_2	U.S. Mid Cap Equity	2026-05-28	1	-0.3%	0.02	0.74
XRT	cluster_2	U.S. Mid Cap Equity	2026-05-28	1	-1.9%	0.12	0.66
XHB	cluster_2	U.S. Mid Cap Equity	2026-05-29	0	0.0%	0.28	0.19
PKB	cluster_2	U.S. Mid Cap Equity	2026-05-28	1	-0.9%	0.79	-0.28
EWT	cluster_8	Emerging Markets Equity	2026-04-08	36	34.6%	0.00	1.00
EEMV	cluster_8	Emerging Markets Equity	2026-04-10	34	11.4%	0.00	1.00
EWY	cluster_8	Emerging Markets Equity	2026-04-09	35	47.8%	0.00	1.00
DEM	cluster_8	Emerging Markets Equity	2026-04-09	35	7.8%	0.00	1.00
AIA	cluster_8	Emerging Markets Equity	2026-04-10	34	24.5%	0.01	0.99
XSOE	cluster_8	Emerging Markets Equity	2026-04-10	34	14.4%	0.03	0.97
GEM	cluster_8	Emerging Markets Equity	2026-04-10	34	12.5%	0.05	0.95
EMGF	cluster_8	Emerging Markets Equity	2026-04-10	34	13.9%	0.05	0.95

ESGE	cluster_8	Emerging Markets Equity	2026-04-13	33	12.5%	0.06	0.94
AAXJ	cluster_8	Emerging Markets Equity	2026-04-13	33	15.1%	0.06	0.94
EEM	cluster_8	Emerging Markets Equity	2026-04-10	34	13.3%	0.06	0.94
EEMA	cluster_8	Emerging Markets Equity	2026-04-13	33	14.8%	0.07	0.93
IEMG	cluster_8	Emerging Markets Equity	2026-04-10	34	12.4%	0.07	0.93
SPEM	cluster_8	Emerging Markets Equity	2026-04-13	33	5.3%	0.25	0.75
FEMS	cluster_8	Emerging Markets Equity	2026-05-28	1	-0.4%	0.79	-0.30
NURE	cluster_15	Real Estate	2026-04-13	33	6.8%	0.01	0.99
SCHH	cluster_15	Real Estate	2026-04-09	35	4.0%	0.08	0.92
RWR	cluster_15	Real Estate	2026-04-10	34	3.8%	0.08	0.92
PSR	cluster_15	Real Estate	2026-04-09	35	4.7%	0.08	0.92
USRT	cluster_15	Real Estate	2026-04-09	35	4.6%	0.08	0.92
ICF	cluster_15	Real Estate	2026-04-08	36	5.0%	0.16	0.84
REET	cluster_15	Real Estate	2026-04-13	33	2.8%	0.21	0.79
REZ	cluster_15	Real Estate	2026-04-14	32	1.9%	0.44	0.56
SHM	cluster_16	Municipal Bond	2026-05-28	1	0.0%	0.02	0.97
SUB	cluster_16	Municipal Bond	2026-05-27	2	0.1%	0.03	0.95
HYD	cluster_16	Municipal Bond	2026-05-22	4	1.2%	0.09	0.91
HYMB	cluster_16	Municipal Bond	2026-05-26	3	0.6%	0.12	0.87
ITM	cluster_16	Municipal Bond	2026-05-28	1	0.2%	0.15	0.82
VTEB	cluster_16	Municipal Bond	2026-05-28	1	0.2%	0.21	0.70
MUB	cluster_16	Municipal Bond	2026-05-28	1	0.1%	0.25	0.65
TFI	cluster_16	Municipal Bond	2026-05-28	1	0.1%	0.24	0.64
CMF	cluster_16	Municipal Bond	2026-05-28	1	0.2%	0.26	0.62
MUNI	cluster_16	Municipal Bond	2026-05-28	1	0.1%	0.29	0.59
NYF	cluster_16	Municipal Bond	2026-05-28	1	0.2%	0.36	0.51
PZA	cluster_16	Municipal Bond	2026-05-29	0	0.0%	0.57	-0.04
FMB	cluster_16	Municipal Bond	2026-05-29	0	0.0%	0.76	-0.19
IAI	cluster_14	Financials	2026-04-10	34	7.5%	0.03	0.97
IAT	cluster_14	Financials	2026-05-20	6	0.9%	0.24	0.65
XLFI	cluster_14	Financials	2026-04-13	33	-0.2%	0.40	0.60
FNCL	cluster_14	Financials	2026-04-13	33	-0.1%	0.46	0.53
KCE	cluster_14	Financials	2026-04-13	33	3.6%	0.92	0.08
KIE	cluster_14	Financials	2026-05-22	4	-4.3%	0.77	-0.52
HYEM	cluster_11	High Yield Bond	2026-04-13	33	1.5%	0.17	0.83
VWOB	cluster_11	High Yield Bond	2026-05-26	3	0.9%	0.17	0.82
PCY	cluster_11	High Yield Bond	2026-05-28	1	0.4%	0.45	0.38
FPE	cluster_11	High Yield Bond	2026-05-27	2	0.2%	1.00	-0.53
IAGG	cluster_5	Aggregate Fixed Income	2026-05-22	4	0.5%	0.00	1.00
BNDX	cluster_5	Aggregate Fixed Income	2026-05-26	3	0.2%	0.00	1.00
SPLB	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.6%	0.02	0.96
VCLT	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.5%	0.02	0.95
IGLB	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.5%	0.02	0.95
IGSB	cluster_5	Aggregate Fixed Income	2026-05-26	3	0.2%	0.08	0.91
USIG	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.3%	0.05	0.90

FLCO	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.3%	0.05	0.90
LQD	cluster_5	Aggregate Fixed Income	2026-05-28	1	0.1%	0.05	0.87
VCSH	cluster_5	Aggregate Fixed Income	2026-05-26	3	0.2%	0.11	0.87
CORP	cluster_5	Aggregate Fixed Income	2026-05-28	1	0.1%	0.08	0.85
QLTA	cluster_5	Aggregate Fixed Income	2026-05-28	1	0.1%	0.05	0.84
SPIB	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.2%	0.12	0.79
SCHO	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.1%	0.19	0.76
ISTB	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.2%	0.16	0.74
IGIB	cluster_5	Aggregate Fixed Income	2026-05-28	1	0.2%	0.14	0.70
SHY	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.1%	0.22	0.70
VGSH	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.1%	0.23	0.69
VCIT	cluster_5	Aggregate Fixed Income	2026-05-28	1	0.1%	0.15	0.67
SPTS	cluster_5	Aggregate Fixed Income	2026-05-27	2	0.1%	0.25	0.66
BLV	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.08	0.65
IUSB	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.15	0.58
VMBS	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.14	0.56
MBB	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.15	0.50
AGG	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.16	0.48
BND	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.16	0.47
SPAB	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.15	0.47
SCHZ	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.17	0.43
BSV	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.26	0.43
BOND	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.20	0.42
SPMB	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.16	0.40
GNMA	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.31	0.22
AGZ	cluster_5	Aggregate Fixed Income	2026-05-11	13	0.1%	0.26	0.04
CMBS	cluster_5	Aggregate Fixed Income	2026-05-29	0	0.0%	0.58	-0.04
SLX	cluster_10	Materials	2026-04-09	35	13.0%	0.04	0.96
DBB	cluster_10	Materials	2026-04-10	34	7.0%	0.11	0.89
COPX	cluster_10	Materials	2026-05-08	14	2.5%	0.11	0.88
XME	cluster_10	Materials	2026-05-26	3	2.1%	0.12	0.87
IYM	cluster_10	Materials	2026-05-26	3	-0.3%	0.23	0.71
VAW	cluster_10	Materials	2026-05-28	1	-0.3%	0.39	0.31
NORW	cluster_10	Materials	2025-12-09	117	31.3%	0.57	0.28
XLB	cluster_10	Materials	2026-05-28	1	-0.4%	0.56	-0.01
MLPX	cluster_10	Materials	2026-04-30	20	-5.2%	0.79	-0.05
RSPM	cluster_10	Materials	2026-05-29	0	0.0%	0.56	-0.13
EMLP	cluster_10	Materials	2026-05-14	10	-2.8%	0.80	-0.38
PID	cluster_9	Asia-Pacific Equity	2026-04-30	20	1.1%	0.00	1.00
EWL	cluster_9	Asia-Pacific Equity	2026-04-15	31	2.4%	0.00	1.00
NFRA	cluster_9	Asia-Pacific Equity	2026-04-09	35	0.7%	0.01	0.99
EWC	cluster_9	Asia-Pacific Equity	2026-04-09	35	4.2%	0.05	0.95
ENZL	cluster_9	Asia-Pacific Equity	2026-05-22	4	5.0%	0.08	0.92
DWX	cluster_9	Asia-Pacific Equity	2026-05-04	18	1.5%	0.18	0.82
EMLC	cluster_9	Asia-Pacific Equity	2026-05-27	2	0.6%	0.22	0.67

EWI	cluster_9	Asia-Pacific Equity	2026-05-05	17	1.1%	0.29	0.66
ECH	cluster_9	Asia-Pacific Equity	2026-05-28	1	0.0%	0.07	0.62
EBND	cluster_9	Asia-Pacific Equity	2026-05-28	1	0.1%	0.35	0.40
EFAV	cluster_9	Asia-Pacific Equity	2026-05-20	6	-1.4%	0.38	-0.07
EWA	cluster_9	Asia-Pacific Equity	2026-05-29	0	0.0%	0.70	-0.21
IGF	cluster_9	Asia-Pacific Equity	2026-05-26	3	-1.6%	0.63	-0.34
LTPZ	cluster_21	Inflation-Protected Fix...	2026-05-28	1	0.0%	0.07	0.85
SPIP	cluster_21	Inflation-Protected Fix...	2026-05-27	2	0.3%	0.59	0.29
SCHP	cluster_21	Inflation-Protected Fix...	2026-05-27	2	0.3%	0.61	0.28
TIP	cluster_21	Inflation-Protected Fix...	2026-05-28	1	0.0%	0.68	0.18
TIPX	cluster_21	Inflation-Protected Fix...	2026-05-27	2	0.3%	0.95	-0.21
TDTF	cluster_21	Inflation-Protected Fix...	2026-05-28	1	0.1%	0.97	-0.38
IGOV	cluster_25	International Fixed Inc...	2026-05-27	2	0.7%	0.37	0.50
FXB	cluster_25	International Fixed Inc...	2026-05-22	4	0.2%	0.93	-0.54
SLV	cluster_17	Gold	2026-05-27	2	1.2%	0.80	-0.19
SIVR	cluster_17	Gold	2026-05-27	2	1.2%	0.81	-0.19
IYK	cluster_18	Consumer Staples	2026-05-06	16	-0.5%	0.52	0.48
KXI	cluster_18	Consumer Staples	2026-05-06	16	-2.2%	0.87	0.11
CNXT	cluster_13	China Equity	2026-04-10	34	19.7%	0.01	0.99
CQQQ	cluster_13	China Equity	2026-04-20	28	4.2%	0.40	0.60
UUP	cluster_26	Leveraged / Inverse	2026-05-18	8	-0.1%	0.00	0.99

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
QQEW	cluster_6	U.S. Large Cap Growth	1.00	1.00
CIBR	cluster_7	U.S. Extended Market Equi...	1.00	1.00
MINT	cluster_24	Cash / Ultra-Short Fixed ...	1.00	1.00
FLOT	cluster_23	Bank Loan / Floating Rate	1.00	1.00
XLV	cluster_12	Healthcare	1.00	1.00
IBDR	cluster_5	Aggregate Fixed Income	1.00	1.00
QUS	cluster_4	U.S. Large Cap Equity	1.00	1.00
ILCV	cluster_1	U.S. Large Cap Value	1.00	1.00
DBEF	cluster_3	Developed ex-US Equity	1.00	1.00
FTXR	cluster_2	U.S. Mid Cap Equity	1.00	1.00
PID	cluster_9	Asia-Pacific Equity	1.00	1.00
FLJH	cluster_22	Japan Equity	1.00	1.00
EWT	cluster_8	Emerging Markets Equity	1.00	1.00
CNXT	cluster_13	China Equity	1.00	0.99
NURE	cluster_15	Real Estate	1.00	0.99
SHM	cluster_16	Municipal Bond	0.99	0.97
IAI	cluster_14	Financials	1.00	0.97
SLX	cluster_10	Materials	1.00	0.96
JNK	cluster_11	High Yield Bond	1.00	0.96
LTPZ	cluster_21	Inflation-Protected Fixed...	0.92	0.85

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
EIDO	cluster_19	India Equity	1.00	-1.00
CHIQ	cluster_13	China Equity	1.00	-1.00
FXC	cluster_9	Asia-Pacific Equity	1.00	-1.00
MOO	cluster_10	Materials	1.00	-1.00
ESPO	cluster_4	U.S. Large Cap Equity	1.00	-1.00
PGF	cluster_11	High Yield Bond	1.00	-0.99

SQQQ	cluster_26	Leveraged / Inverse	0.99	-0.99
SDIV	cluster_8	Emerging Markets Equity	1.00	-0.98
PBJ	cluster_18	Consumer Staples	0.98	-0.98
GSG	cluster_20	Broad Commodities	1.00	-0.98
REM	cluster_1	U.S. Large Cap Value	0.98	-0.97
BKLN	cluster_23	Bank Loan / Floating Rate	1.00	-0.96
GLD	cluster_17	Gold	0.96	-0.94
FXE	cluster_25	International Fixed Income	0.97	-0.89
LMBS	cluster_5	Aggregate Fixed Income	0.92	-0.77
IPAY	cluster_7	U.S. Extended Market Equi...	0.93	-0.65
KIE	cluster_14	Financials	0.77	-0.52
TDTF	cluster_21	Inflation-Protected Fixed...	0.97	-0.38
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.97	-0.36
PKB	cluster_2	U.S. Mid Cap Equity	0.79	-0.28

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_19	11	FLAT	0.303	0.094	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.490	0.077	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.688	0.042	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.731	0.094	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.608	0.220	0.00035	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.750	0.135	0.00030	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_20	10	FLAT	0.596	0.170	0.00040	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.822	0.024	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.822	0.092	0.00022	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.642	0.190	0.00047	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.767	0.170	0.00039	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.706	0.190	0.00043	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_19	11	FLAT	0.497	0.216	0.00055	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.767	0.180	0.00043	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	FLAT	0.844	0.127	0.00034	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.818	0.202	0.00048	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	LONG	0.877	0.162	0.00030	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.836	0.236	0.00046	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_19	11	FLAT	0.494	0.323	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	LONG	0.878	0.250	0.00045	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_19	11	FLAT	0.445	0.331	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.878	0.264	0.00052	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.805	0.269	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_10	21	LONG	0.750	0.358	0.00084	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_19	11	FLAT	0.501	0.367	0.00105	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.855	0.255	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_19	11	FLAT	0.573	0.412	0.00107	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.331	0.00071	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	LONG	0.904	0.241	0.00063	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	LONG	0.919	0.227	0.00056	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20